Neural Network Approximation of the LHCb Magnetic Field Map for Fast Track Extrapolation

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Abstract

Track extrapolation through the LHCb dipole magnet requires repeated evaluation of a trilinear-interpolated magnetic field map—a memory-bound operation that dominates the cost of Runge–Kutta-based extrapolators. We investigate replacing the field-map lookup with a small neural network that maps spatial coordinates $(x, y, z) \to (B_x, B_y, B_z)$. A first generation of 30 models, spanning five widths, three depths, and two activations, achieved mean absolute errors as low as 0.5 G but suffered from catastrophic failure in strong-field regions: relative errors exceeding 99% where $|\mathbf{B}| > 100\,\mathrm{G}$. Analysis reveals that the extreme class imbalance of the field distribution—89.6% of the grid volume has $|\mathbf{B}| < 1\,G$—causes standard MSE training to collapse the network toward zero output. This motivates a second generation of experiments comparing eight loss functions and curriculum-learning schedules designed to correct this failure mode.

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1 Motivation

1.1 Track extrapolation in LHCb

The LHCb experiment at the Large Hadron Collider reconstructs the trajectories of charged particles produced in proton–proton collisions. A central task in the reconstruction software is *track extrapolation*: propagating the five-parameter track state $(x, y, t_x, t_y, q/p)$ from one z-plane to another through the magnetic field of the dipole magnet. Multiple extrapolation algorithms are implemented in the `Rec` stack, the most physics-accurate being the fourth-order Runge–Kutta (RK4) extrapolator (`TrackRungeKuttaExtrapolator`).

Each RK4 step requires four evaluations of the magnetic field $\mathbf{B}(x, y, z)$; a typical track traversing the full magnet ($z \in [200, 9500]$ mm) uses on the order of 40 field evaluations. High-level trigger (HLT) processing at 30 MHz crossing rate makes extrapolation a significant fraction of the total CPU budget, and any factor-of-two speedup would be directly beneficial to physics throughput.

1.2 The field-map lookup bottleneck

The LHCb magnetic field is stored as a regular 3D grid of $81 \times 81 \times 146 = 957{,}906$ vertices spanning

$$\begin{aligned} x &\in [-4000, 4000]\ \text{mm}, \quad \Delta x = 100\,mm, \\ y &\in [-4000, 4000]\ \text{mm}, \quad \Delta y = 100\,mm, \\ z &\in [-500, 14000]\ \text{mm}, \quad \Delta z = 100\,mm. \end{aligned} \tag{1}$$

Each vertex stores the three components (B_x, B_y, B_z) in Gauss. At query time, trilinear interpolation is performed over the eight surrounding grid corners, requiring eight loads from a $957{,}906 \times 3 \times 8 = 23\,MB$ look-up table (single precision). This data structure far exceeds the L1 cache (32 KB) of modern CPUs, making each field evaluation a cache-miss-dominated, memory-bound operation.

1.3 Replacing the field map with a neural network

The key observation is that the field map is a smooth, deterministic function $\mathbf{B} : \mathbb{R}^3 \to \mathbb{R}^3$. A small feedforward network (MLP) with $\mathcal{O}(200\text{–}20{,}000)$ parameters would fit entirely in L1 cache and replace every trilinear interpolation with a sequence of fused multiply-add operations—a compute-bound workload that modern CPUs handle efficiently.

The extrapolation algorithm itself (RK4, Cash–Karp, parabolic, etc.) remains unchanged; only the field evaluation is substituted. This is the minimal-invasiveness approach: no retraining of the integrator, no change to the track model, and a single header file (`FieldMapNNWeights.h`) encapsulates the entire replacement.

The equations of motion for a charged particle in a magnetic field are

$$\frac{d\mathbf{r}}{dz} = \begin{pmatrix} t_x \\ t_y \\ 1 \end{pmatrix}, \qquad \frac{dt_x}{dz} = \frac{q}{p}\, n\big(t_y(t_x B_x + B_z) - (1 + t_x^2)\, B_y\big), \tag{2}$$

with an analogous expression for dt_y/dz, where $n = c\sqrt{1 + t_x^2 + t_y^2}$ is a normalisation factor and c is the speed of light. The field components B_x, B_y, B_z enter linearly, so any proportional error in the field translates directly into a proportional error in the computed deflection. The *relative* accuracy of the field approximation is therefore the physically meaningful metric, not the mean absolute error.

1.4 Computational cost breakdown

To quantify the potential speedup, we performed native C++ micro-benchmarks of every sub-component of a Cash–Karp track extrapolation (see companion document `extrapolator_cost_analysis.pdf` for full details). All timings use `std::chrono::high_resolution_clock` with 100,000 iterations on an AMD EPYC 7551P, compiled at `-O2 -march=native`.

Key measured timings.

Component	Time (ns)	Notes
Trilinear interpolation	100	8 loads from 23 MB grid
NN [32] SiLU (227 params)	872	8.8× slower (32 `exp()` calls)
Lorentz derivative	14	14 FLOPs + 1 `sqrt`
Cash–Karp step (trilinear)	385	6 field + 6 deriv + bookkeeping
Full track, 10 steps (trilinear)	2625	≈ 2.50 μs production
Full track, 10 steps (NN [32])	30,269	11.5× *slower*

FLOP decomposition. A single Cash–Karp step comprises 558 FLOPs: B-field lookup accounts for 60% (336 FLOPs), the Lorentz derivative for 18% (102 FLOPs), and Butcher-tableau bookkeeping for 22% (120 FLOPs).

Two speedup strategies. **Strategy A** (replace only the field lookup) is bounded by Amdahl's law with $f = 0.60$ and a maximum speedup of 2.5×. However, the NN [32] SiLU is 8.8× *slower* per call than trilinear, making Strategy A non-viable with current activations: the measured full-track slowdown is 11.5×. **Strategy B** (replace the entire extrapolator with a single NN forward pass) eliminates all 60 field evaluations and is estimated at ∼2.0 µs—a 1.3× speedup over the 2.5 µs production baseline.

These results motivate pursuing Strategy B (full NN extrapolator) as the primary path, while Strategy A requires faster activations (ReLU) or hardware acceleration to break even. Nevertheless, an NN field map with sufficient accuracy remains valuable for physics validation and as a stepping stone toward a full NN extrapolator.

2 Generation 1: Architecture Grid Search

2.1 Experimental setup

A grid search was conducted over 30 network architectures, defined by the Cartesian product of:

- **Width**: $\{32, 64, 128, 256, 512\}$ hidden units per layer,
- **Depth**: $\{1, 2, 3\}$ hidden layers,
- **Activation**: $\{\text{ReLU}, \text{SiLU}\}$.

All models share the same `FieldMLP` architecture: a Sequential stack of `Linear`+activation layers with a 3-dimensional input (x, y, z) and 3-dimensional output (B_x, B_y, B_z). Inputs and outputs are independently standardised (zero mean, unit variance).

Training used all 957,906 grid vertices as training data and 100,000 random off-grid points (with trilinear-interpolated ground truth) for validation. The loss function was the standard mean squared error (MSE) on the normalised targets:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|^2 , \tag{3}$$

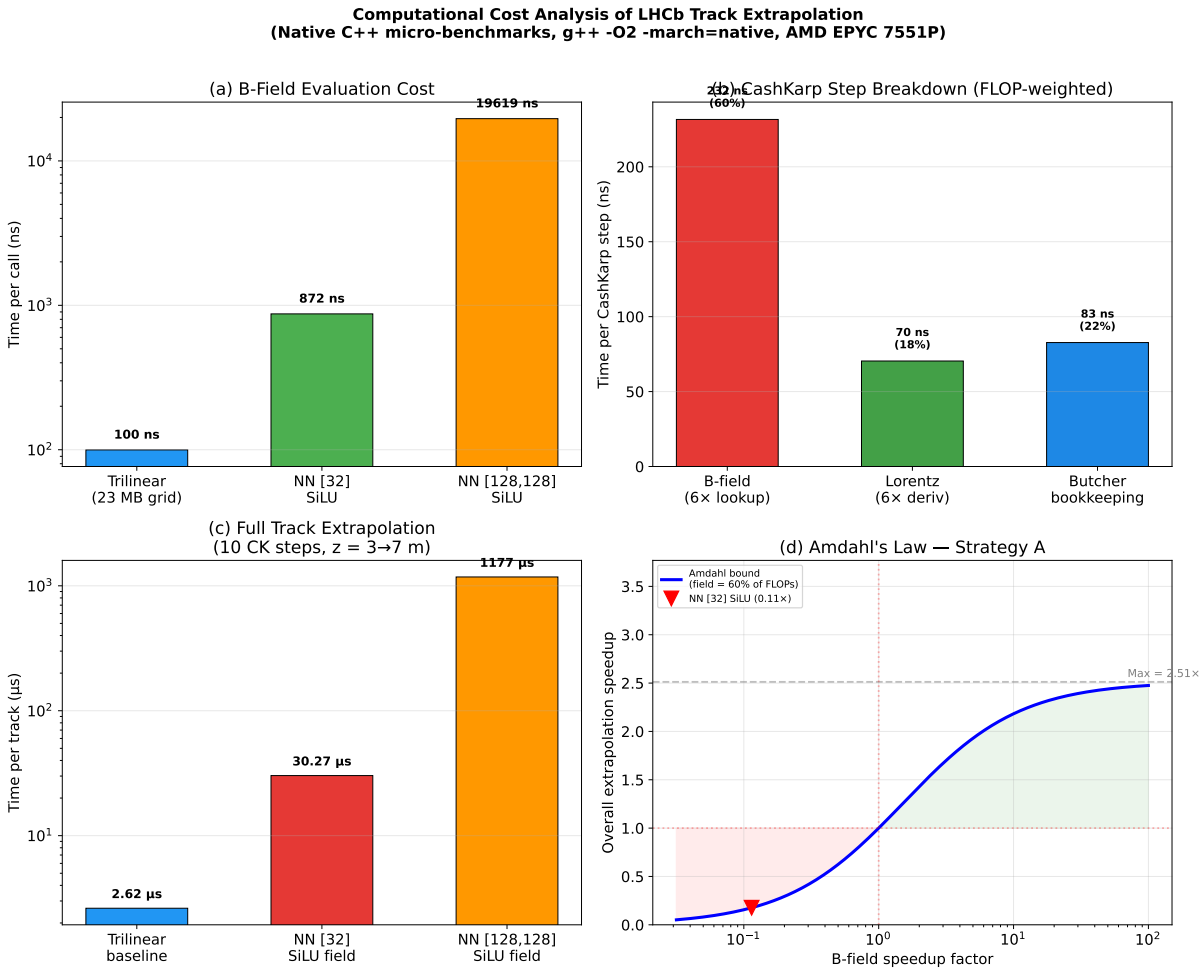

Figure 1: Computational cost breakdown of LHCb track extrapolation from native C++ micro-benchmarks. (a) Per-call B-field evaluation cost. (b) FLOP-weighted Cash–Karp step decomposition. (c) Full track timing with different field methods. (d) Amdahl's law bound for Strategy A, showing the NN [32] SiLU operating point deep in the "slower" regime.

Table 1: Summary of Generation 1 results. MAE and maximum error are in Gauss. SiLU consistently outperforms ReLU at every architecture.

Activation	Depth×Width	Parameters	FLOPs	MAE (G)	Max error (G)
SiLU	3×256	133,379	268,288	0.50	14,675
SiLU	2×128	17,411	35,328	0.70	16,988
SiLU	1×32	227	512	0.82	16,814
ReLU	2×512	266,243	531,456	0.60	17,398
ReLU	1×32	227	320	1.02	16,995

where $\hat{\mathbf{y}}_i$ and $\mathbf{y}_i$ are the predicted and true normalised field vectors. Training ran for up to 200 epochs with Adam (learning rate 10^{-3}), cosine annealing, and early stopping (patience = 20).

All 30 jobs were submitted to the Nikhef STBC HTCondor cluster with one GPU per job.

2.2 Results

Table 1 presents a subset of the 30 models. The global MAE ranged from 0.50 G (SiLU 3×256) to 1.15 G (ReLU 1×32), and SiLU was uniformly superior to ReLU at every width and depth.

At first glance these errors appear excellent: a 0.5 G MAE against a peak field of 33024 G (3.30 T) suggests a mean relative error of $\sim 10^{-5}$. However, inspection of the *maximum* error column reveals that every model has a maximum error of approximately 17000 G—roughly half the peak field strength. This maximum does not improve meaningfully with model capacity: the 227-parameter model and the 133,379-parameter model both show maximum errors of ∼15000–17000 G.

2.3 Failure analysis: the class-imbalance catastrophe

The root cause is the extreme imbalance of the field magnitude distribution over the grid volume. Table 2 quantifies this.

Table 2: Distribution of $|\mathbf{B}|$ over the 957,906 field-map grid vertices.

Region	Fraction of grid
$\lvert\mathbf{B}\rvert < 1\,G$	89.6%
$\lvert\mathbf{B}\rvert < 10\,G$	99.0%
$\lvert\mathbf{B}\rvert > 100\,G$	0.1%
$\lvert\mathbf{B}\rvert > 1000\,G$	$< 0.1\%$

The LHCb dipole field is concentrated in a narrow cylindrical core around the beam axis near $z \approx 5000\,mm$, while the vast majority of the grid volume is in field-free regions upstream, downstream, and off-axis. With MSE training (Eq. 3), a sample at $|\mathbf{B}| = 1\,G$ and a sample at $|\mathbf{B}| = 33000\,G$ contribute equally to the loss if both have the same *absolute* residual. Since 99% of samples have $|\mathbf{B}| < 10\,G$, the optimal strategy under uniform MSE is to predict approximately zero everywhere. This explains why:

1. The *global* MAE is dominated by the 89.6% of points with near-zero field, producing a deceptively low value (∼0.5 G).

2. The *maximum* error is approximately equal to the peak field (∼17000 G), because the network has learned to output ≈ 0 even in the magnet core.

3. Increasing model capacity (from 227 to 133,379 parameters) does not help, because the loss landscape itself provides no gradient signal to learn the strong-field region.

We verified this diagnosis by computing relative errors at specific high-field points. At the field maximum $(x, y, z) = (0, -3900, 5000)$ mm, where the true field is $B_y = -33024\,G$, the best-MAE model (SiLU 3×256) predicts $B_y \approx -1936\,G$—a relative error of 94%. The speed-viable model (SiLU 1×32) predicts $B_y \approx 3\,G$—a relative error of $> 99.99\%$.

These errors propagate directly through Eq. 2: a track at 1 GeV/c momentum extrapolated from $z = 200$ to $9500\,mm$ using the NN field (viable model) arrives with a position offset of 1503 mm and a slope error of 0.33 compared to the trilinear reference. The NN field map is, in its current form, physically unusable.

3 Generation 2: Loss Function and Training Strategy Comparison

3.1 Motivation

The Generation 1 failure is not a capacity problem but a *loss-function* problem. The field distribution spans five orders of magnitude (∼0.01 G to ∼33000 G), and any loss that treats all residuals equally will be dominated by the 99% of near-zero samples. To fix this, the loss must either:

(a) **Re-weight** samples so that the rare high-field points contribute proportionally more to the gradient;

(b) **Transform** the targets into a space where the dynamic range is compressed (e.g. logarithmic); or

(c) Use a **relative** error metric that penalises proportional deviations regardless of magnitude.

3.2 Experimental design

We fix the architecture to SiLU 2×128 (17,411 parameters, 35,328 FLOPs) and compare eight training strategies. All other hyperparameters are held constant: 300 epochs, Adam with cosine annealing, batch size 4096, patience 30, identical train/validation splits.

Table 3: The eight training strategies compared in Generation 2. B_i denotes the field magnitude of sample i in Gauss; $B_{\text{ref}} = 10\,G$.

Name	Loss / schedule	Category
`baseline`	$\frac{1}{N}\sum\|\hat{\mathbf{y}}_i - \mathbf{y}_i\|^2$ (normalised)	Control
`weighted_mse`	$\frac{1}{N}\sum w_i\,\|\hat{\mathbf{y}}_i - \mathbf{y}_i\|^2, \quad w_i = 1 + \|B_i\|/B_{\text{ref}}$	Re-weight (a)
`weighted_strong`	Same, $w_i = 1 + (\|B_i\|/B_{\text{ref}})^2$	Re-weight (a)
`relative_mse`	$\frac{1}{N}\sum\left\|\frac{\hat{\mathbf{B}}_i - \mathbf{B}_i}{\|\mathbf{B}_i\| + \epsilon}\right\|^2, \quad \epsilon = 1\,G$	Relative (c)
`log_space`	MSE on $\text{sign}(B) \cdot \log(\|B\| + 1)$	Transform (b)
`mixed`	$0.5\,\mathcal{L}_{\text{MSE}} + 0.5\,\mathcal{L}_{\text{rel}}$	Hybrid (a+c)
`curriculum_lin`	Phase 1 (30%): uniform MSE; Phase 2: w_i ramps linearly $1 \to 50$	Curriculum + (a)
`curriculum_exp`	Phase 1 (30%): uniform MSE;Phase 2: w_i ramps exponentially $1 \to 50$	Curriculum + (a)

The curriculum strategies use a two-phase schedule. In Phase 1 (the first 30% of epochs), the model trains with uniform MSE to learn the spatial structure of the field. In Phase 2,

importance weights based on $|\mathbf{B}|$ are gradually introduced—linearly or exponentially—up to a maximum weight factor of 50, forcing the network to allocate capacity to the strong-field region.

3.3 Evaluation metrics

In addition to the standard MAE and maximum error (in Gauss), we now track *relative* error metrics computed over validation points where $|\mathbf{B}| > 1\,G$:

- **Relative MAE**: $\langle |B_{\text{pred}} - B_{\text{true}}|/|B_{\text{true}}| \rangle$
- **Relative p95**: 95th percentile of $|B_{\text{pred}} - B_{\text{true}}|/|B_{\text{true}}|$
- **Relative max**: worst-case relative error

A successful strategy must achieve relative MAE $\lesssim 1\%$ across the full field range to be considered for deployment, corresponding to trajectory errors of $\mathcal{O}(\text{mm})$ rather than $\mathcal{O}(\text{m})$.

3.4 Results

Results will be added after the eight training jobs complete on the Nikhef STBC HTCondor cluster.